Xiaoling Hu

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Current Position

- **Harvard Medical School, Harvard University, USA** Aug. 2023 - Present
  Postdoctoral Research Fellow
  - Hosted by Prof. Juan Eugenio Iglesias and Prof. Bruce Fischl

Research Interests

My research interests lie in machine learning, computer vision, with applications in healthcare, multimodal AI, generative AI, VLM/LLMs, and trustworthy learning systems. In particular, I am interested in:

- **Topological Deep Learning**
- **Trustworthy Machine Learning**
- **Multimodal AI and Generative AI (GenAI)**
- **VLM/LLM Reasoning, Efficiency and Confidence**

Education

- **Department of CS, Stony Brook University, USA** Jan. 2018 - June 2023
  Doctor of Philosophy
- **Department of EE, Tsinghua University, China** Sep. 2014 - June 2017
  Master of Engineering
- **Department of EE, Huazhong University of Science and Technology, China** Sep. 2010 - June 2014
  Bachelor of Science

Selected Honors and Awards

- CVPR Oral (Top 1%) 2025
- Catacosinos Fellowship 2023
- ICLR Spotlight (Top 5%) 2023
- ECCV Oral (Top 2%) 2022
- ICLR Spotlight (Top 5%) 2021
- NeurIPS Travel Award 2019

Professional Service

Area Chair or Senior Program Committee

- Conference on Neural Information Processing Systems (NeurIPS) 2025, 2026
- International Conference on Learning Representations (ICLR) 2027
- The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026
- International Conference on Artificial Intelligence and Statistics (AISTATS) 2026
- AAAI Conference on Artificial Intelligence 2027
- The IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2027
- International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI) 2025, 2026

Selected Publications (* indicates equal contribution, __ denotes students (co-)mentored by me, ‡ denotes (co)-senior supervision)

[1] ***RankByGene*: Gene-Guided Histopathology Representation Learning Through Cross-Modal Ranking Consistency**
Wentao Huang, Meilong Xu, **Xiaoling Hu**, Shahira Abousamra, Aniruddha Ganguly, Saarthak Kapse, Alisa Yurovsky, Prateek Prasanna, Tahsin Kurc, Joel Saltz, Michael L. Miller, Chao Chen
*IEEE Transactions on Medical Imaging (**TMI**)*, 2026

[2] **Multi-Channel Uncertainty-Weighted Score Matching for Conditional Diffusion in Medical UDA**
Chen Li, Meilong Xu, **Xiaoling Hu**, Weimin Lyu, Chao Chen
*European Conference on Computer Vision (**ECCV**)*, 2026

[3] **Act Like a Pathologist: Tissue-Aware Whole Slide Image Reasoning**
Wentao Huang, Weimin Lyu, Peiliang Lou, Qingqiao Hu, **Xiaoling Hu**, Shahira Abousamra, Wenchao Han, Ruifeng Guo, Jiawei Zhou, Chao Chen, Chen Wang
*The IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**)*, 2026

[4] **MATCH: Multi-faceted Adaptive Topo-Consistency for Semi-Supervised Histopathology Segmentation**
Meilong Xu*, **Xiaoling Hu***, Shahira Abousamra, Chen Li, Chao Chen
*Thirty-Ninth Conference on Neural Information Processing Systems (**NeurIPS**)*, 2025
*Short version is selected as **Oral Presentation** at Imageomics Workshop at NeurIPS 2025*

[5] ***Learn2Synth*: Learning Optimal Data Synthesis Using Hypergradients for Brain Image Segmentation**
Xiaoling Hu, Xiangrui Zeng, Oula Puonti, Juan Eugenio Iglesias, Bruce Fischl‡, Yaël Balbastre‡
*International Conference on Computer Vision (**ICCV**)*, 2025

[6] ***TopoCellGen*: Generating Histopathology Cell Topology with a Diffusion Model**
Meilong Xu, Saumya Gupta, **Xiaoling Hu**, Chen Li, Shahira Abousamra, Dimitris Samaras, Prateek Prasanna, Chao Chen
*The IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**)*, 2025 (**Oral Presentation, acceptance rate $<$ 1%**)
Short version is also presented at CVPR 2025 Workshop GMCV

[7] **Hierarchical Uncertainty Estimation for Learning-based Registration in Neuroimaging**
Xiaoling Hu, Karthik Gopinath, Peirong Liu, Malte Hoffmann, Koen Van Leemput, Oula Puonti‡, Juan Eugenio Iglesias‡
*International Conference on Learning Representations (**ICLR**)*, 2025

[8] **Semi-supervised Segmentation of Histopathology Images with Noise-Aware Topological Consistency**
Meilong Xu, **Xiaoling Hu**, Saumya Gupta, Shahira Abousamra, Chao Chen
*European Conference on Computer Vision (**ECCV**)*, 2024

[9] **Brain-ID: Learning Contrast-agnostic Anatomical Representations for Brain Imaging**
Peirong Liu, Oula Puonti, **Xiaoling Hu**, Daniel C. Alexander, Juan Eugenio Iglesias
*European Conference on Computer Vision (**ECCV**)*, 2024

[10] **Anomaly-guided weakly supervised lesion segmentation on retinal OCT images**
Jiaqi Yang, Nitish Mehta, Gozde Merve Demirci, **Xiaoling Hu**, Meera Ramakrishnan, Mina Naguib, Chao Chen, Chialing Tsai
*Medical Image Analysis (**MedIA**)*, 2024

[11] **Topology-Aware Uncertainty for Image Segmentation**
Saumya Gupta, Yikai Zhang, **Xiaoling Hu**, Prateek Prasanna, Chao Chen
*Thirty-Seventh Conference on Neural Information Processing Systems (**NeurIPS**)*, 2023

[12] **Calibrating Uncertainty for Semi-Supervised Crowd Counting**
Chen Li, **Xiaoling Hu**, Shahira Abousamra, Chao Chen
*International Conference on Computer Vision (**ICCV**)*, 2023

[13] **Enhancing Modality-Agnostic Representations via Meta-Learning for Brain Tumor Segmentation**
Aishik Konwer, **Xiaoling Hu**, Xuan Xu, Joseph Bae, Chao Chen, Prateek Prasanna
*International Conference on Computer Vision (**ICCV**)*, 2023

[14] **Learning Probabilistic Topological Representations Using Discrete Morse Theory**
Xiaoling Hu, Dimitris Samaras, Chao Chen
*International Conference on Learning Representations (**ICLR**)*, 2023 (**Spotlight Presentation, notable-top-25% of accepted papers**)
*Short version is selected as **Oral Presentation** at Medical Imaging meets NeurIPS Workshop, 2023*

[15] **Confidence Estimation Using Unlabeled Data**
Chen Li, **Xiaoling Hu**, Chao Chen
*International Conference on Learning Representations (**ICLR**)*, 2023

[16] **Structure-Aware Image Segmentation with Homotopy Warping**
Xiaoling Hu
*Thirty-Sixth Conference on Neural Information Processing Systems (**NeurIPS**)*, 2022

[17] **Learning Topological Interactions for Multi-Class Medical Image Segmentation**
Saumya Gupta*, **Xiaoling Hu***, James Kaan, Michael Jin, Mutshipay Mpoy, Katherine Chung, Gagandeep Singh, Mary Saltz, Tahsin Kurc, Joel Saltz, Apostolos Tassiopoulos, Prateek Prasanna, Chao Chen
*European Conference on Computer Vision (**ECCV**)*, 2022 (**Oral Presentation, acceptance rate 2.7%**)

[18] **Trigger Hunting with a Topological Prior for Trojan Detection**
Xiaoling Hu, Xiao Lin, Michael Cogswell, Yi Yao, Susmit Jha, Chao Chen
*International Conference on Learning Representations (**ICLR**)*, 2022

[19] **Topology-Aware Segmentation Using Discrete Morse Theory**
Xiaoling Hu, Yusu Wang, Li Fuxin, Dimitris Samaras, Chao Chen
*International Conference on Learning Representations (**ICLR**)*, 2021 (**Spotlight Presentation, acceptance rate 5.6%**)

[20] **Topology-Preserving Deep Image Segmentation**
Xiaoling Hu, Li Fuxin, Dimitris Samaras, Chao Chen
*Thirty-Third Conference on Neural Information Processing Systems (**NeurIPS**)*, 2019

Preprints

[1] **ORCE: Order-Aware Alignment of Verbalized Confidence in Large Language Models**
Chen Li, **Xiaoling Hu**, Songzhu Zheng, Jiawei Zhou, Chao Chen
Tech Report, 2026

[2] **Topo-R1: Detecting Topological Anomalies via Vision-Language Models**
Meilong Xu, Qingqiao Hu, **Xiaoling Hu**, Shahira Abousamra, Xin Yu, Weimin Lyu, Kehan Qi, Dimitris Samaras, Chao Chen
Tech Report, 2026

[3] **Unrolled Networks are Conditional Probability Flows in MRI Reconstruction**
Kehan Qi, Saumya Gupta, **Xiaoling Hu**, Qingqiao Hu, Weimin Lyu, Chao Chen
Tech Report, 2025

[4] **LoC-Path: Learning to Compress for Pathology Multimodal Large Language Models**
Qingqiao Hu, Weimin Lyu, Meilong Xu, Kehan Qi, **Xiaoling Hu**, Saumya Gupta, Jiawei Zhou, Chao Chen
Tech Report, 2025

[5] **RB-FT: Rationale-Bootstrapped Fine-Tuning for Video Classification**
Meilong Xu, Di Fu, Jiaxing Zhang, Gong Yu, Jiayu Zheng, **Xiaoling Hu**, Dongdi Zhao, Feiyang Li, Chao Chen, Yong Cao
Tech Report, 2025

[6] **Uncertainty Estimation for Pretrained Medical Image Registration Models via Transformation Equivariance**
Lin Tian, **Xiaoling Hu**, Juan Eugenio Iglesias
Tech Report, 2025

[7] **Learning to Upscale 3D Segmentations in Neuroimaging**
Xiaoling Hu, Peirong Liu, Dina Zemlyanker, Jonathan Williams Ramirez, Oula Puonti, Juan Eugenio Iglesias
Tech Report, 2025

Mentoring

- Chen Li (**ICLR'23, ICCV'23, MICCAI'24, ECCV'26**), Ph.D. student at Department of BMI, Stony Brook University
  Since Fall 2021
- Meilong Xu (**ECCV'24, CVPR'25 Oral Presentation, NeurIPS'25**), Ph.D. student at Department of CS, Stony Brook University
  Since Summer 2023
- Wentao Huang (**MICCAI'24**, **CVPR'26**, **TMI'26**), Ph.D. student at Department of CS, Stony Brook University
  Since Summer 2023
- Qingqiao Hu, Ph.D. student at Department of CS, Stony Brook University
  Since Fall 2024
- Kehan Qi, Ph.D. student at Department of BMI, Stony Brook University
  Since Fall 2025
- Jiaqi Yang (**MICCAI'21, ISBI'21, MedIA'24, JBHI'25**), Ph.D. student at Department of CS, CUNY
  Spring 2020 – Summer 2023

- Saumya Gupta (**ECCV'22 Oral Presentation, NeurIPS'23**), Ph.D. student at Department of CS, Stony Brook University
  Fall 2021 – Summer 2023
- Mahmudul Hasan (**MICCAI'24**), Ph.D. student at Department of CS, Stony Brook University
  Summer 2023 – Summer 2024
- Luca Drole, Master student in BME, ETH Zurich
  Spring - Fall 2025
- John Xie, High School student → University of Michigan
  Summer 2021

Talks

Principled Learning for Medical AI: Structure, Reliability, and Interpretability

- School of Computing & SCI Institute, University of Utah — Mar. 2026
- School of Computing, University of Georgia — Mar. 2026
- Department of EECS, University of Tennessee, Knoxville — Feb. 2026
- Department of CS, Missouri S&T — Jan. 2026
- HIT Webinar - *Healthcare, Intelligence, Technology* — Aug. 2025
- Department of CSE, University of North Texas — Nov. 2025

***Learn2Synth*: A Learnable Data Synthesis Strategy for Image Segmentation**

- Nobrainer Seminar, Massachusetts Institute of Technology — June 2024

Deep Structural Reasoning for Biomedical Imaging

- School of CAI, Arizona State University — Feb. 2024

Topology-Aware Deep Image Segmentation

- MICCAI'23 tutorial on *Topology-Driven Image Analysis*, Vancouver — Oct. 2023

Learning Topological Representations for Deep Image Understanding

- Department of CS, Florida State University — Apr. 2023
- Department of BMI, Ohio State University — Mar. 2023
- Department of CS, Rochester Institute of Technology — Feb. 2023
- Department of ECE, University of California, Riverside — Feb. 2023
- Athinoula A. Martinos Center for Biomedical Imaging, Massachusetts General Hospital & Harvard Medical School — Nov. 2022

Learning Probabilistic Topological Representations Using Discrete Morse Theory

- Medical Imaging meets NeurIPS Workshop, New Orleans — Dec. 2022

Topology-Informed Image Analysis

- Center for Computational Neuroscience, Flatiron Institute — Oct. 2022

Topology-Aware Deep Image Segmentation

- Geometry and Topology meet Data Analysis and Machine Learning — Aug. 2021

Topology-aware Segmentation Using Discrete Morse Theory

- International Conference on Learning Representations (ICLR) — May 2021

Industry Experiences

- **Allen Institute, USA** — May 2022 - Aug. 2022
  Research Intern
  Mentor: *Dr.* Matheus Viana
  Topic: Topology-Aware Image Segmentation
- **United Imaging Intelligence (UII), USA** — May 2021 - Aug. 2021
  Research Intern
  Mentor: *Dr.* Shanhui Sun
  Topic: Deep Shape Model Based Network
- **Tencent Youtu Lab, China** — Jun. 2017 - Jan. 2018
  Research Intern
  Mentor: *Dr.* Yuwing Tai
  Topic: Clothes Detection, Attribute Prediction